

## ESPERANCE, Australia

WMO: 946380

Lat: 33.8300S

Lon: 121.8925E

Elev: 89

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 43.2       | 44.6 | 29.8                          | 23.9 | 66.5 | 33.0 | 27.6 | 62.8 | 28.5                  | 58.4 | 25.7 | 59.2 | 9.2                   | 10   | 0.562 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 15.3                   | 92.7            | 64.5 | 86.7 | 63.9 | 81.8 | 63.6 | 69.2                | 80.2 | 67.8 | 77.4 | 66.7 | 75.4 | 14.7                 | 10   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 66.3                            | 97.4 | 71.1 | 65.1 | 93.0 | 70.6 | 63.9 | 89.1 | 70.0 | 33.4          | 80.7 | 32.2 | 77.4 | 31.4 | 75.3 | 78.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 24.9          | 22.6 | 20.7 | DB                         | 39.8 | 105.9              | 1.5 | 3.4                                                | 38.7 | 108.4      | 37.9 | 110.4      | 37.0 | 112.3      | 36.0 | 114.7 | (4) |
| (5)               |      |      | WB                         | 38.1 | 72.4               | 1.4 | 2.4                                                | 37.1 | 74.2       | 36.2 | 75.6       | 35.4 | 76.9       | 34.4 | 78.7  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 61.8   | 68.5  | 68.9  | 67.2  | 64.3  | 60.1  | 56.3  | 54.8  | 55.7  | 57.6  | 60.1  | 63.3  | 65.8  | (6)  |  |
|                                                                        | DBStd       | 6.84   | 5.32  | 5.00  | 5.27  | 5.10  | 4.32  | 3.29  | 3.27  | 4.04  | 4.48  | 5.29  | 5.64  | 5.45  | (7)  |  |
|                                                                        | HDD50       | 5      | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 1     | 1     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD65       | 1709   | 13    | 9     | 30    | 74    | 165   | 262   | 316   | 292   | 228   | 175   | 98    | 47    | (9)  |  |
|                                                                        | CDD50       | 4325   | 574   | 529   | 534   | 429   | 312   | 189   | 151   | 177   | 228   | 314   | 400   | 488   | (10) |  |
|                                                                        | CDD65       | 555    | 123   | 118   | 99    | 52    | 13    | 1     | 0     | 2     | 6     | 23    | 48    | 70    | (11) |  |
|                                                                        | CDH74       | 3724   | 796   | 649   | 593   | 319   | 77    | 3     | 1     | 31    | 76    | 235   | 415   | 529   | (12) |  |
|                                                                        | CDH80       | 1476   | 344   | 258   | 232   | 101   | 13    | 0     | 0     | 6     | 19    | 90    | 179   | 234   | (13) |  |
| (14) Wind                                                              | WSAvg       | 11.4   | 12.6  | 12.2  | 11.1  | 10.0  | 10.0  | 10.9  | 11.3  | 11.1  | 11.6  | 11.6  | 12.2  | 12.3  | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 24.1   | 0.7   | 1.0   | 1.2   | 1.8   | 2.8   | 3.0   | 3.8   | 3.3   | 2.4   | 1.9   | 1.3   | 0.7   | (15) |  |
|                                                                        | PrecMax     | 34.2   | 3.5   | 5.7   | 5.8   | 6.1   | 7.3   | 7.6   | 7.6   | 5.9   | 4.7   | 4.6   | 3.5   | 2.4   | (16) |  |
|                                                                        | PrecMin     | 17.6   | 0.0   | 0.0   | 0.0   | 0.1   | 0.6   | 1.1   | 0.9   | 1.1   | 0.6   | 0.2   | 0.0   | 0.0   | (17) |  |
|                                                                        | PrecStd     | 4.2    | 0.7   | 1.1   | 1.3   | 1.2   | 1.5   | 1.3   | 1.4   | 1.2   | 1.1   | 1.0   | 0.8   | 0.6   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 101.6 | 97.0  | 95.3  | 89.6  | 81.6  | 72.4  | 71.8  | 78.2  | 81.9  | 90.2  | 94.5  | 97.9  | (19) |  |
|                                                                        |             | MCWB   | 67.3  | 66.9  | 65.4  | 62.1  | 60.0  | 56.0  | 55.5  | 57.4  | 58.2  | 59.8  | 62.6  | 64.6  | (20) |  |
|                                                                        | 2%          | DB     | 88.9  | 87.8  | 86.3  | 82.1  | 75.4  | 68.8  | 67.4  | 71.2  | 74.6  | 80.9  | 84.8  | 86.0  | (21) |  |
|                                                                        |             | MCWB   | 66.3  | 66.5  | 64.3  | 61.6  | 58.8  | 55.0  | 54.3  | 55.7  | 57.2  | 59.4  | 61.5  | 64.1  | (22) |  |
|                                                                        | 5%          | DB     | 81.6  | 81.2  | 80.2  | 76.9  | 71.5  | 66.1  | 64.6  | 67.2  | 69.7  | 73.6  | 77.3  | 78.8  | (23) |  |
|                                                                        |             | MCWB   | 66.1  | 66.2  | 64.1  | 61.7  | 58.2  | 55.0  | 53.6  | 54.8  | 56.5  | 59.0  | 61.5  | 64.0  | (24) |  |
|                                                                        | 10%         | DB     | 76.8  | 76.6  | 75.4  | 72.6  | 68.2  | 63.6  | 62.0  | 63.8  | 65.9  | 69.0  | 72.5  | 74.0  | (25) |  |
|                                                                        |             | MCWB   | 65.0  | 65.7  | 64.1  | 61.3  | 57.8  | 54.2  | 53.0  | 54.0  | 55.7  | 58.1  | 61.1  | 62.7  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 71.3  | 72.2  | 69.2  | 66.8  | 64.6  | 60.4  | 58.5  | 59.9  | 61.7  | 64.1  | 67.1  | 68.5  | (27) |  |
|                                                                        |             | MCDB   | 84.6  | 82.7  | 79.0  | 74.8  | 71.8  | 66.6  | 64.7  | 69.1  | 72.7  | 74.8  | 79.4  | 81.6  | (28) |  |
|                                                                        | 2%          | WB     | 69.1  | 69.5  | 67.5  | 65.3  | 62.4  | 58.6  | 56.9  | 58.2  | 59.6  | 61.9  | 64.8  | 66.5  | (29) |  |
|                                                                        |             | MCDB   | 81.6  | 79.1  | 75.9  | 72.4  | 68.2  | 63.8  | 62.7  | 65.9  | 67.8  | 72.2  | 75.6  | 78.1  | (30) |  |
|                                                                        | 5%          | WB     | 67.4  | 68.0  | 66.5  | 64.1  | 60.8  | 57.0  | 55.5  | 56.7  | 58.2  | 60.3  | 63.3  | 65.0  | (31) |  |
|                                                                        |             | MCDB   | 77.6  | 76.6  | 74.5  | 70.9  | 67.3  | 62.4  | 61.5  | 63.8  | 65.9  | 69.8  | 72.7  | 75.1  | (32) |  |
|                                                                        | 10%         | WB     | 66.1  | 66.7  | 65.5  | 63.0  | 59.3  | 55.6  | 54.2  | 55.2  | 56.9  | 59.1  | 61.8  | 63.6  | (33) |  |
|                                                                        |             | MCDB   | 74.8  | 74.3  | 73.0  | 69.7  | 65.6  | 61.4  | 60.0  | 61.8  | 64.2  | 67.5  | 70.5  | 72.4  | (34) |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 15.8  | 15.3  | 15.8  | 16.5  | 15.6  | 14.7  | 14.5  | 15.8  | 16.4  | 16.6  | 16.5  | 15.4  | (35) |  |
|                                                                        | 5% DB       | MCDBR  | 27.4  | 26.3  | 25.9  | 24.1  | 21.3  | 19.0  | 19.3  | 22.5  | 24.3  | 27.4  | 28.2  | 26.7  | (36) |  |
|                                                                        |             | MCWBR  | 10.2  | 9.7   | 9.3   | 9.7   | 9.6   | 9.9   | 9.9   | 10.8  | 10.8  | 11.5  | 10.9  | 10.7  | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 21.5  | 19.4  | 18.4  | 17.2  | 16.3  | 15.1  | 15.7  | 18.6  | 19.9  | 21.4  | 22.4  | 21.7  | (38) |  |
|                                                                        |             | MCWBR  | 9.3   | 8.4   | 8.2   | 8.9   | 9.4   | 9.6   | 9.4   | 10.2  | 10.0  | 10.3  | 9.8   | 9.8   | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.360 | 0.354 | 0.336 | 0.326 | 0.308 | 0.299 | 0.299 | 0.309 | 0.323 | 0.339 | 0.344 | 0.342 | (40) |  |
|                                                                        | taud        |        | 2.527 | 2.554 | 2.594 | 2.615 | 2.635 | 2.630 | 2.626 | 2.589 | 2.544 | 2.516 | 2.526 | 2.554 | (41) |  |
|                                                                        | Ebn at Noon |        | 310   | 305   | 300   | 285   | 275   | 269   | 275   | 287   | 299   | 307   | 313   | 317   | (42) |  |
|                                                                        | Edh at Noon |        | 35    | 33    | 30    | 26    | 23    | 21    | 22    | 26    | 30    | 34    | 35    | 35    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2317  | 2019  | 1664  | 1253  | 923   | 788   | 854   | 1142  | 1532  | 1906  | 2186  | 2393  | (44) |  |
| (45)                                                                   | RadStd      |        | 126   | 139   | 110   | 62    | 54    | 29    | 45    | 55    | 61    | 117   | 142   | 144   | (45) |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |  | +0.40 | N/A     | -1.22  | N/A     | N/A   | N/A   | N/A         | -108  | +167  | +79   |
| (47) Regional (6 neighbors) |  | +0.41 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -108  | +167  | +79   |

Nomenclature: See separate page